0. Introduction

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SS26 MTH344: Mathematical Statistics

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Motivating Example

- This course aims to cover advanced topics in mathematical statistics, focusing primarily on **statistical inference**.
- Statistical inference is the process of identifying the most plausible probabilistic models to describe observed data and evaluating their validity.
- To motivate our discussion, let's consider the *Juryeonggu*, an ancient Korean die.

Left: A replica at the Gyeongju National Museum. Right: Inscriptions on each face recorded at the time of excavation.

Motivating Example

- *Juryeonggu* (酒令具; AD668 $\sim$ AD935) is an ancient die used for drinking games during a historical dynasty in what is now South Korea.
- It has 14 faces, six square (□) and eight hexagonal (⬡), with each face assigning a penalty, e.g., 三盞一去: Drink three cups at once.

Motivating Example

- **Q:** This die features two distinct types of faces: □ and ⬡. What outcomes can we expect when we roll it? Are these two types of faces equally likely to occur?

Motivating Example

- A direct approach is to define a 14-dimensional probability vector $\boldsymbol{p} = (p_1, \ldots, p_{14})^T \in \{\boldsymbol{p} : p_i \geq 0, \sum_{i=1}^{14} p_i = 1\}$ representing the probabilities of each face.

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	p_1	p_2	p_3	p_4	p_5	p_6	p_7

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	p_8	p_9	p_{10}	p_{11}	p_{12}	p_{13}	p_{14}

- Each $\boldsymbol{p}$ identifies a distinct probability distribution.
- For example, $\boldsymbol{p} = (1/14, \ldots, 1/14)^T$ represents the uniform model, where all fourteen faces are equally likely to occur.

Motivating Example

- By assuming that faces of the same type are equally likely, we can parameterize the distribution using a single scalar $p \in [0, 1/6]$, where p denotes the probability of observing a specific $\square$ face.

Outcome	1st $\square$	2nd $\square$	3rd $\square$	4th $\square$	5th $\square$	6th $\square$	1st $\hexagon$
Probability	p	p	p	p	p	p	$\frac{1-6p}{8}$

Outcome	2nd $\hexagon$	3rd $\hexagon$	4th $\hexagon$	5th $\hexagon$	6th $\hexagon$	7th $\hexagon$	8th $\hexagon$
Probability	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$	$\frac{1-6p}{8}$

- For example, $p = 1/14$ recovers the uniform distribution, and $p = 1/7$ indicates that a $\square$ face is eight times more likely to appear than a $\hexagon$ face.
- **Q:** What would be the optimal choice of p to best describe the true data-generating process of the die?

Motivating Example: Estimation

- The relative frequencies provide direct estimates for the probabilities of each of the fourteen individual outcomes.
- Wang et al. (2005) conducted an experiment rolling the die $n = 7,000$ times using two different materials: wood and Fiberglass Reinforced Plastic (FRP).
- The observed relative frequencies for the wooden die are as follows:

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Relative Frequency	$\frac{523}{7000}$	$\frac{533}{7000}$	$\frac{477}{7000}$	$\frac{525}{7000}$	$\frac{518}{7000}$	$\frac{492}{7000}$	$\frac{456}{7000}$

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Relative Frequency	$\frac{512}{7000}$	$\frac{506}{7000}$	$\frac{488}{7000}$	$\frac{505}{7000}$	$\frac{487}{7000}$	$\frac{482}{7000}$	$\frac{496}{7000}$

- This yields the empirical estimate $\hat{\boldsymbol{p}} = \left(\frac{523}{7000}, \ldots, \frac{496}{7000}\right)^T$.

Motivating Example: Estimation

- Under the single-parameter model, we might estimate p using only the first □ face:

$$\hat{p} = \frac{523}{7000} \approx 7.5\% \tag{1}$$

- Alternatively, we can pool the observations from all square faces to obtain a more robust estimate:

$$\hat{p} = \left(\frac{523 + 533 + 477 + 525 + 518 + 492}{7000} \right) \frac{1}{6} \approx 7.3\% \tag{2}$$

- Using this pooled estimate, the fitted distribution is given by:

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	7.3%	7.3%	7.3%	7.3%	7.3%	7.3%	7.0%

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%

Motivating Example: Hypothesis Testing

- Based on the pooled estimate, the probabilities of observing a specific □ face and a specific ⬡ face are estimated to be $\hat{p} \approx 7.3\%$ and $(1 - 6\hat{p})/8 \approx 7.0\%$, respectively.
- We might want to test the conjecture (or *hypothesis*) that the die is perfectly fair:

$$p = 1/14. \tag{3}$$

- **Q**: How can we formally determine if the data are consistent with this hypothesis?

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Relative Frequency	$\frac{523}{7000}$	$\frac{533}{7000}$	$\frac{477}{7000}$	$\frac{525}{7000}$	$\frac{518}{7000}$	$\frac{492}{7000}$	$\frac{456}{7000}$

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Relative Frequency	$\frac{512}{7000}$	$\frac{506}{7000}$	$\frac{488}{7000}$	$\frac{505}{7000}$	$\frac{487}{7000}$	$\frac{482}{7000}$	$\frac{496}{7000}$

- **Q**: Furthermore, how can we test the underlying assumption that faces of the same shape are equally likely, i.e.,

$$p_1 = \cdots = p_6 \quad \text{and} \quad p_7 = \cdots = p_{14}? \tag{4}$$

Lecture Topics

- Through the motivating example, we briefly introduced the three core pillars of statistical inference:

  (i) Probabilistic Models, (ii) Estimation, and (iii) Hypothesis Testing.

- This course consists of three modules corresponding directly to these pillars:

  M1. Review of Probability Theory
  M2. Sampling Distributions and Estimation
  M3. Hypothesis Testing

- The course content is primarily based on the textbook, **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007).
- However, it also incorporates substantial supplementary and advanced material beyond the text.

Lecture Topics

M1. Review of Probability Theory

1. Probability
2. Random Variables
3. Joint Distributions
4. Expected Values

M2. Sampling Distributions and Estimation

5. Limit Theorems
6. Distributions Derived from the Normal Distribution
8. Estimation of Parameters and Fitting of Probability Distributions

M3. Hypothesis Testing

9. Testing Hypotheses and Assessing Goodness of Fit
11. Comparing Two Samples
12. The Analysis of Variance

References I

Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.

Wang, M.-O., Seo, J.-C., and Lim, I.-K. (2005). Probability research of wooden die for drinking game as ethnic custom mathematics. *Journal for History of Mathematics*, 18(4):67–84.